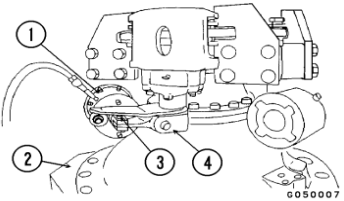
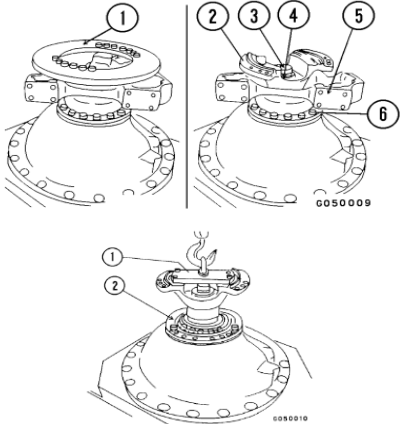
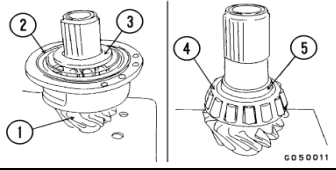
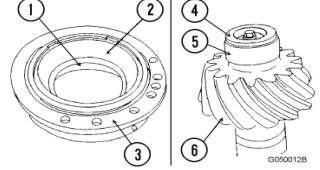
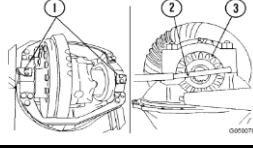
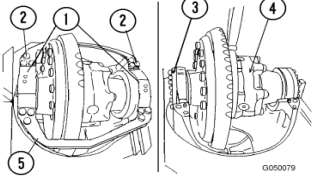
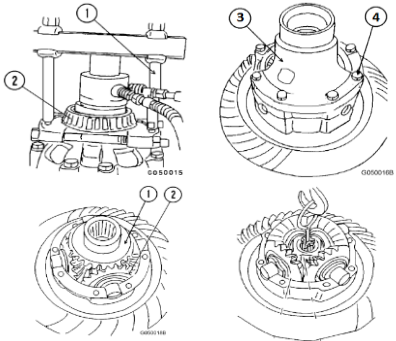
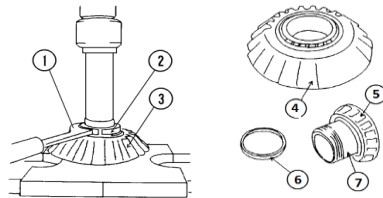
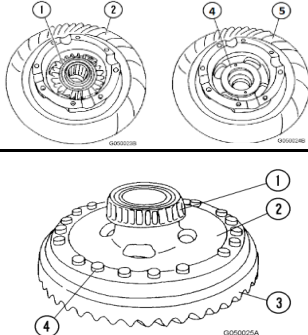


DISASSEMBLY CHECK SHEET		WO NUMBER		DOCUMENT NO			
DIFFERENTIAL		COMP.MODEL		PUBLIC DATE			
HD 1500-7		COMP.S/N		REVISION NO			
		UNIT MODEL		PAGE			
PROCESS		START		DISASSEMBLY BY			
		FINISH					
NO	PART TO REMOVE OR INSPECT	CONDITIONS TO BE INSPECTION	MECH	GL	SKETCH DRAWING OR PHOTOES	REMARK	
1		⚠ IMPORTANT ⚠ Set the Differential on stand and check worn abnormally, crack, chipping, scratch, scoring and threat of component. Using a portable power source, pump hydraulic oil into the park brake spring cylinder to retract the rod, then pull out connecting pin (3) of adjuster lever (4), and remove spring cylinder			 NOTE: The HD1500 has three brake caliper assemblies.		
2		- Remove all three parking brake assemblies - Remove retaining capscrew (3), and mounting capscrews (6). Then, remove coupling (2) together with holder (4) and the O-ring. - Remove support (5). - Install coupling and lifting device (1) and screw in pusher bolts, then lift off pinion and carrier assembly (2). - Check the number and thickness of the shims, and keep together in a safe place <table border="1" style="margin-left: auto; margin-right: auto;"> <tr> <td>Shims</td> </tr> </table>	Shims				
Shims							
3	PINION ASSEMBLY	- Remove carrier (2) together with outer bearing (3) from pinion gear (1) - Remove spacer (5), then remove inner bearing (4)					
		- Remove holder (4), then remove center bearing inner race (5) from pinion gear (6) - Remove bearing outer races (2) and (1) from carrier (3).					
4	DIFFERENTIAL GEAR	- Remove locks (1) - loosen the left and right side bearing adjustment nuts (3) until they can be turned by hand.					
		- Remove capscrews (2). Remove plates (5) and caps (1). - Lift off differential gear assembly (4) - Remove left and right side bearing adjustment nuts (3).					

DISASSEMBLY CHECK SHEET DIFFERENTIAL HD 1500-7	WO NUMBER		DOCUMENT NO	
	COMP.MODEL		PUBLIC DATE	
	COMP.S/N		REVISION NO	
	UNIT MODEL		PAGE	

NO	PART TO REMOVE OR INSPECT	CONDITIONS TO BE INSPECTED	MECH	GL	SKETCH DRAWING OR PHOTOS	REMARK
5	DIFFERENTIAL GEAR	- Using a special tool to remove bearing (2). - Punch mark the case sections and remove mounting capscrews (4). Remove case (3). - Remove thrust washer from case. - Remove side gear (1). - Remove pinion gear assembly (1, Figure 5-16) together with cross shaft (2).				
6	PINION GEAR ASSY	- Hold pinion gear assembly (3) with a press. Use wrench (1) to remove ring nut (2). - Push out shaft (7) with the press, and remove pinion (4), bearing (5), and collar (6). - Remove the bearing from the shaft - Remove bearing outer races, from pinion gear.				
7	BEVEL GEAR ASSY	- Remove side gear (1) - Remove thrust washer (4) - Remove bevel gear (3) from case (2). - Remove bearing (1) from the case. - Remove snap ring. Remove bearing (1) from differential case (2).				

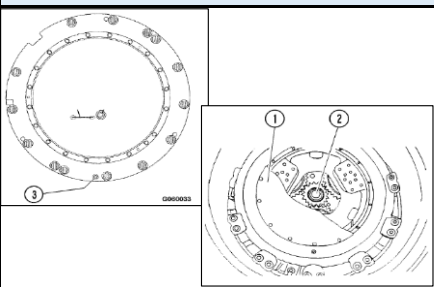
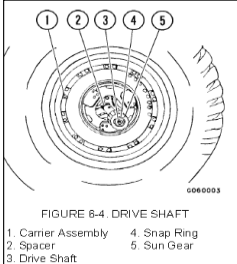
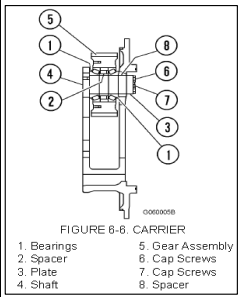
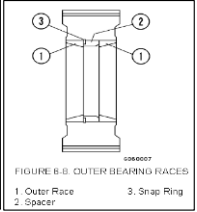
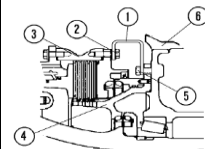
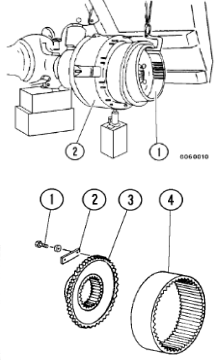
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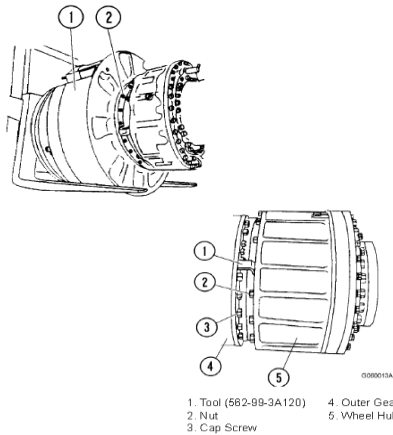
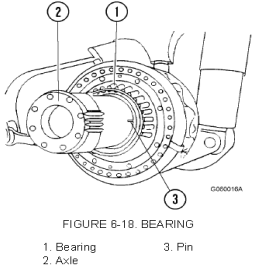
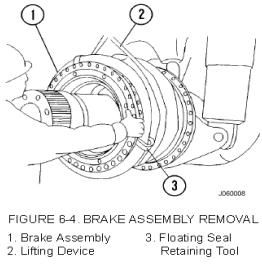
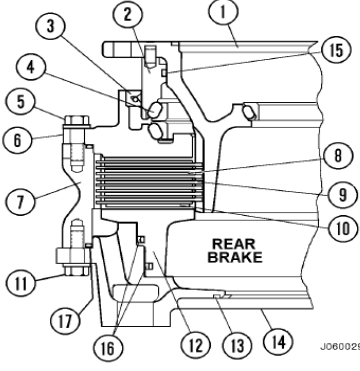
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GROUP LEADER (GL)

DISASSEMBLY CHECK SHEET		WO NUMBER	DOCUMENT NO			
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		UNIT MODEL	PAGE			
PROCESS	START FINISH	DISASSEMBLY BY				
NO	PART TO REMOVE OR INSPECT	CONDITIONS TO BE INSPECTED	MECH GL SKETCH DRAWING OR PHOTOS REMARK			
1	SETTING OF REAR AXLE ASSEMBLY	⚠ IMPORTANT ⚠ Set the rear axle assembly vertically on stand and check worn abnormally, crack, chipping, scratch, scoring and threat of component. Rear axle assembly: 13209 kg - Remove drain plug (3) and drain the oil from the final drive case. Oil capacity is 120 l - Remove bolt(1) that secure cover. Remove the cover(2). - Set rear axle assembly (3) with planetary carrier (4)				
2	PLANETARY CARRIER ASSEMBLY	⚠ IMPORTANT ⚠ Set the rear axle assembly vertically and check worn abnormally, crack, chipping, scratch, scoring and threat of component. - Install an eyebolt at the end of drive shaft (3). Use a bar to pull the drive shaft from the wheel. Attach a lifting device to the shaft for support as it is pulled from the carrier. - Remove snap ring (4) from the drive shaft. - Remove sun gear (5). - Remove spacer (2). - Attach appropriate lifting apparatus to carrier assembly (1). Remove nine cap screws (6, Figure 6-6) and six cap screws (7). Remove plate (3). Use a press to remove shaft (4). Remove gear assembly (5). Remove two bearings (1) and spacer (2). Remove outer race (1, Figure 6-8) and spacer(2) ⚠ IMPORTANT ⚠ NOTE: The bearing is an adjustment-free bearing. Keep the bearing together as a set, in a safe place. - Remove snap ring (3).Figure 6-8) - Remove spacer (8, Figure 6-6) from the carrier.	 FIGURE 6-4. DRIVE SHAFT 1. Carrier Assembly 4. Snap Ring 2. Spacer 5. Sun Gear 3. Drive Shaft  FIGURE 6-6. CARRIER 1. Bearings 5. Gear Assembly 2. Spacer 6. Cap Screws 3. Plate 7. Cap Screws 4. Shaft 8. Spacer  FIGURE 6-8. OUTER BEARING RACES 1. Outer Race 3. Snap Ring 2. Spacer			
3	RING GEAR ASSEMBLY	- Remove the retainer and shims. Use three fixture tools (562-99-3A120) (1, Figure 6-10) to secure disc brake inner gear (4) to outer gear (3). To install the tools, remove three cap screws (2) from the outer gear, and insert cap screws (5) into the three tool mounting tapped holes in the inner gear. - Sling ring gear temporarily and remove retainer (5) - Remove shims (6) - a check the quatinty and thickness of the shims <table border="1"> <tr> <td>Actual</td> </tr> <tr> <td> </td> </tr> </table> - Disassemble the ring gear assembly as follows. Remove mounting bolt(1) then remove holder (2). Remove inner hub (3) from ring gear (4).	Actual		  	
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4	REAR WHEEL HUB ASSEMBLY	⚠ IMPORTANT ⚠ Attach an appropriate lifting sling to rear wheel hub assembly (1) or support with a forklift. The weight of the assembly is approximately 968 kg (2134 lbs). - Remove nuts (2, Figure 6-15), except those on the seal retainer tools - Remove wheel hub assembly (2, Figure 6-16) slowly until the outer bearing is unseated. Remove outer bearing (1) and bearing inner race pin. The weight of the bearing is approximately 76 kg (168 lb). c. Remove the wheel hub assembly. d. Remove outer races from the wheel hub assembly			 1. Tool (562-99-3A120) 4. Outer Gear 2. Nut 5. Wheel Hub 3. Cap Screw	
5	INNER BEARING	- Remove inner bearing (1, Figure 6-18 and bearing inner race pin (3). The weight of the bearing is approximately 75 kg (165 lbs). 8. Remove floating seal carrier (13, Figure 6-1) from the axle.			 FIGURE 6-18. BEARING 1. Bearing 3. Pin 2. Axle	
5	BRAKE ASSEMBLY	⚠ IMPORTANT ⚠ Check worn abnormally, crack, chipping, scratch, scoring and threat of component. brake assembly : 1100 Kg - Remove capscrews and washers (17, Figure 6-5). Carefully lift the brake assembly outward until it is clear of the rear axle. - Move the brake assembly to a clean work area for disassembly.			 FIGURE 6-4. BRAKE ASSEMBLY REMOVAL 1. Brake Assembly 3. Floating Seal Retaining Tool 2. Lifting Device	
6	REAR BRAKE ASSEMBLY	NOTE: Match mark brake assembly components to ensure correct orientation of parts during reassembly. - Position the brake assembly on a work surface, as shown in Figure 6-7. - Install a suitable lifting device on inner gear (1, Figure 6-7) or on the wheel hub. - Carefully, lift the inner gear from the assembly ⚠ IMPORTANT ⚠ Damage to the floating seal may result when removing the inner gear. Use caution and remove the gear, slowly. - Remove capscrews and flatwashers (5) from hub (6). - Lift the hub from ring gear (7). - Note the order of assembly of the discs. Remove dampers (10), separator plates (8) and friction discs (9). Also note the notches at seven locations on the periphery of the damper plates and separator plates. The discs must be reassembled correctly to ensure proper oil flow.			 FIGURE 6-7. BRAKE ASSEMBLY COMPONENTS 1. Inner Gear 10. Damper 2. Seal Carrier 11. Capscrew & Washer 3. Lip Seal 12. Piston 4. Floating Seal 13. O-Ring 5. Capscrew & Washer 14. Cylinder 6. Hub 15. O-Ring 7. Ring Gear 16. Piston Seals 8. Separator Plate 17. O-Ring 9. Friction Disc	



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PART TO REMOVE OR INSPECT	CONDITIONS TOBE INSPECTION	MECH	GL	SKECTH DRAWING OR PHOTOES	REMARK
	- Remove Bolt and flatwashers (11). Remove ring gear (7) from cylinder (14). - Remove piston (12). <i>NOTE: If piston removal is difficult, plug any open ports at the piston apply pressure passages. Slowly apply air pressure at one port to push the piston out of the cylinder bore.</i> - Remove the two halves of floating seal assembly (4). - Remove and discard all O-ring seals. - Remove lip seal (3) from the bore of hub (6).				

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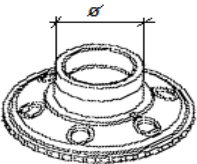
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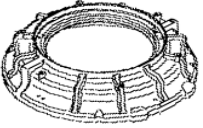
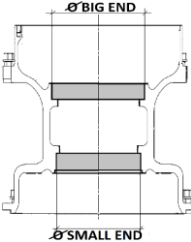
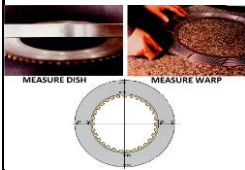
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FINAL DRIVE		COMP.MODEL	PUBLIC DATE
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NO	PART TO BE INSPECTED	CHECKING POINT	SKECTH DRAWING OR PHOTOES	VISUAL CHECK & MEASUREMENT																																	
1	CARRIER HOUSING (P/N B562-22-32611)	- Planetary shaft hole - O - Ring Groove - All round		<table border="1" style="display: inline-table; margin-right: 10px;"> <tr><th>Contens check</th><th>Actual</th></tr> <tr><td>Normal</td><td></td></tr> <tr><td>Crack</td><td></td></tr> <tr><td>Scratch</td><td></td></tr> <tr><td>Broken</td><td></td></tr> <tr><td>Wear</td><td></td></tr> </table> <table border="1" style="display: inline-table;"> <tr><th>U/A</th><th>U/R</th><th>R/N</th></tr> <tr><td></td><td></td><td></td></tr> </table>	Contens check	Actual	Normal		Crack		Scratch		Broken		Wear		U/A	U/R	R/N																		
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2	CARRIER GEAR (P/N B562-22-32531)	- Gear teeth - Bearing fitting portion - Snap ring groove		<table border="1" style="display: inline-table; margin-right: 10px;"> <tr><th>Contens check</th><th>Actual</th></tr> <tr><td>Normal</td><td></td></tr> <tr><td>Crack</td><td></td></tr> <tr><td>Scratch</td><td></td></tr> <tr><td>Broken</td><td></td></tr> <tr><td>Wear</td><td></td></tr> </table> <table border="1" style="display: inline-table;"> <tr><th>U/A</th><th>U/R</th><th>R/N</th></tr> <tr><td></td><td></td><td></td></tr> </table>	Contens check	Actual	Normal		Crack		Scratch		Broken		Wear		U/A	U/R	R/N																		
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3	CARRIER SHAFT (P/N B562-22-32511)	- Bearing fitting portion - Carrier bore fitting portion - Ball fitting portion		<table border="1" style="display: inline-table; margin-right: 10px;"> <tr><th>Contens check</th><th>Actual</th></tr> <tr><td>Normal</td><td></td></tr> <tr><td>Crack</td><td></td></tr> <tr><td>Scratch</td><td></td></tr> <tr><td>Broken</td><td></td></tr> <tr><td>Wear</td><td></td></tr> </table> <table border="1" style="display: inline-table;"> <tr><th>U/A</th><th>U/R</th><th>R/N</th></tr> <tr><td></td><td></td><td></td></tr> </table>	Contens check	Actual	Normal		Crack		Scratch		Broken		Wear		U/A	U/R	R/N																		
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4	HUB (P/N 562-22-32310)	- Internal spline teeth - External gear teeth - Bearing fitting portion - Retainer fitting portion		<table border="1" style="display: inline-table; margin-right: 10px;"> <tr><th>X</th><th>Y</th><th>Z</th></tr> <tr><td></td><td></td><td></td></tr> <tr><td></td><td></td><td></td></tr> <tr><td></td><td></td><td></td></tr> <tr><td></td><td></td><td></td></tr> </table> <table border="1" style="display: inline-table; margin-right: 10px;"> <tr><th>Contens check</th><th>Actual</th></tr> <tr><td>Normal</td><td></td></tr> <tr><td>Crack</td><td></td></tr> <tr><td>Scratch</td><td></td></tr> <tr><td>Broken</td><td></td></tr> <tr><td>Wear</td><td></td></tr> </table> Remark : <table border="1" style="display: inline-table;"> <tr><th>U/A</th><th>U/R</th><th>R/N</th></tr> <tr><td></td><td></td><td></td></tr> </table>	X	Y	Z													Contens check	Actual	Normal		Crack		Scratch		Broken		Wear		U/A	U/R	R/N			
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5	SPACER & SUN GEAR SPACER (P/N 562-22-12432) GEAR (P/N 562-22-12422)	- Gear teeth - Internal spline teeth - Side surface		<table border="1" style="display: inline-table; margin-right: 10px;"> <tr><th>Contens check</th><th>Actual</th></tr> <tr><td>Normal</td><td></td></tr> <tr><td>Crack</td><td></td></tr> <tr><td>Scratch</td><td></td></tr> <tr><td>Broken</td><td></td></tr> <tr><td>Wear</td><td></td></tr> </table> <table border="1" style="display: inline-table;"> <tr><th>U/A</th><th>U/R</th><th>R/N</th></tr> <tr><td></td><td></td><td></td></tr> </table>	Contens check	Actual	Normal		Crack		Scratch		Broken		Wear		U/A	U/R	R/N																		
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6	RETAINER (P/N 562-22-32210)	- Matting Face - All round		<table border="1" style="display: inline-table; margin-right: 10px;"> <tr><th>Contens check</th><th>Actual</th></tr> <tr><td>Normal</td><td></td></tr> <tr><td>Crack</td><td></td></tr> <tr><td>Scratch</td><td></td></tr> <tr><td>Broken</td><td></td></tr> <tr><td>Wear</td><td></td></tr> </table> <table border="1" style="display: inline-table;"> <tr><th>U/A</th><th>U/R</th><th>R/N</th></tr> <tr><td></td><td></td><td></td></tr> </table>	Contens check	Actual	Normal		Crack		Scratch		Broken		Wear		U/A	U/R	R/N																		
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8	CYLINDER (PN 562-33-31110)	- Friction surface - Seal ring contact surface - O-ring groove - All round		<table border="1" style="display: inline-table; margin-right: 10px;"> <tr><th>Contens check</th><th>Actual</th></tr> <tr><td>Normal</td><td></td></tr> <tr><td>Crack</td><td></td></tr> <tr><td>Scratch</td><td></td></tr> <tr><td>Broken</td><td></td></tr> <tr><td>Wear</td><td></td></tr> </table> <table border="1" style="display: inline-table;"> <tr><th>U/A</th><th>U/R</th><th>R/N</th></tr> <tr><td></td><td></td><td></td></tr> </table>	Contens check	Actual	Normal		Crack		Scratch		Broken		Wear		U/A	U/R	R/N																		
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9	PISTON (P/N B562-33-31120)	- Friction surface - seal ring contact surface - seal ring groove - All round		<table border="1" style="display: inline-table; margin-right: 10px;"> <tr><th>Contens check</th><th>Actual</th></tr> <tr><td>Normal</td><td></td></tr> <tr><td>Crack</td><td></td></tr> <tr><td>Scratch</td><td></td></tr> <tr><td>Broken</td><td></td></tr> <tr><td>Wear</td><td></td></tr> </table> <table border="1" style="display: inline-table;"> <tr><th>U/A</th><th>U/R</th><th>R/N</th></tr> <tr><td></td><td></td><td></td></tr> </table>	Contens check	Actual	Normal		Crack		Scratch		Broken		Wear		U/A	U/R	R/N																		
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VISUAL & MEASUREMENT CHECK SHEET FINAL DRIVE HD 1500-7		WO NUMBER	DOCUMENT NO
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11	RETAINER (P/N B562-33-31320)	- Oil seal contact surface - Floating seal contact surface - O-ring groove		<table border="1" style="display: inline-table; margin-right: 10px;"> <tr><th>Contens check</th><th>Actual</th></tr> <tr><td>Normal</td><td></td></tr> <tr><td>Crack</td><td></td></tr> <tr><td>Scratch</td><td></td></tr> <tr><td>Broken</td><td></td></tr> <tr><td>Wear</td><td></td></tr> </table> <table border="1" style="display: inline-table;"> <tr><th>U/A</th><th>U/R</th><th>R/N</th></tr> <tr><td></td><td></td><td></td></tr> </table>	Contens check	Actual	Normal		Crack		Scratch		Broken		Wear		U/A	U/R	R/N																																																																							
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12	RETAINER (P/N 562-33-31220)	- Floating seal contact surface - O-ring groove - All round		<table border="1" style="display: inline-table; margin-right: 10px;"> <tr><th>Contens check</th><th>Actual</th></tr> <tr><td>Normal</td><td></td></tr> <tr><td>Crack</td><td></td></tr> <tr><td>Scratch</td><td></td></tr> <tr><td>Broken</td><td></td></tr> <tr><td>Wear</td><td></td></tr> </table> <table border="1" style="display: inline-table;"> <tr><th>U/A</th><th>U/R</th><th>R/N</th></tr> <tr><td></td><td></td><td></td></tr> </table>	Contens check	Actual	Normal		Crack		Scratch		Broken		Wear		U/A	U/R	R/N																																																																							
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17	HUB (P/N 562-22-32711)	- Bearing fitting portion - thread portion & Matting Face Big end <table border="1" style="display: inline-table; margin-right: 10px;"> <tr><th>Standart</th><th>tolerance</th></tr> <tr><td></td><td></td></tr> </table> Small end <table border="1" style="display: inline-table;"> <tr><th>Standart</th><th>tolerance</th></tr> <tr><td></td><td></td></tr> </table>	Standart	tolerance			Standart	tolerance				<table border="1" style="margin-right: 10px;"> <tr><th colspan="3">Big end</th></tr> <tr><th>X</th><th>Y</th><th>Z</th></tr> <tr><td></td><td></td><td></td></tr> <tr><td></td><td></td><td></td></tr> <tr><td></td><td></td><td></td></tr> </table> <table border="1"> <tr><th colspan="3">Small end</th></tr> <tr><th>X</th><th>Y</th><th>Z</th></tr> <tr><td></td><td></td><td></td></tr> <tr><td></td><td></td><td></td></tr> <tr><td></td><td></td><td></td></tr> </table> <table border="1" style="margin-top: 10px;"> <tr><th>Contens check</th><th>Actual</th></tr> <tr><td>Normal</td><td></td></tr> <tr><td>Crack</td><td></td></tr> <tr><td>Scratch</td><td></td></tr> <tr><td>Broken</td><td></td></tr> <tr><td>Wear</td><td></td></tr> </table> <table border="1" style="margin-top: 10px;"> <tr><th>U/A</th><th>U/R</th><th>R/N</th></tr> <tr><td></td><td></td><td></td></tr> </table>	Big end			X	Y	Z										Small end			X	Y	Z										Contens check	Actual	Normal		Crack		Scratch		Broken		Wear		U/A	U/R	R/N																																	
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18	PLATE (P/N 561-33-71230)	- Deformation - tooth flank wear on external gear - Heat spot or metal transfer THICKNESS <table border="1" style="display: inline-table; margin-right: 10px;"> <tr><th>Standart</th><th>tolerance</th></tr> <tr><td>3</td><td>2,85</td></tr> </table> DISH & WRAP <table border="1" style="display: inline-table;"> <tr><th>Standart</th><th>tolerance</th></tr> <tr><td>0,5</td><td>0,2</td></tr> </table>	Standart	tolerance	3	2,85	Standart	tolerance	0,5	0,2		<table border="1" style="width:100%;"> <tr><th colspan="8">THICKNESS</th></tr> <tr><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th><th>7</th><th>8</th></tr> <tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr> </table> <table border="1" style="width:100%;"> <tr><th colspan="8">DISH & WRAP</th></tr> <tr><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th><th>7</th><th>8</th></tr> <tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr> </table> Remark : <table border="1" style="margin-top: 10px;"> <tr><th>U/A</th><th>U/R</th><th>R/N</th></tr> <tr><td></td><td></td><td></td></tr> </table>	THICKNESS								1	2	3	4	5	6	7	8																									DISH & WRAP								1	2	3	4	5	6	7	8																	U/A	U/R	R/N			
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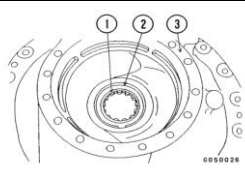
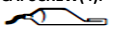
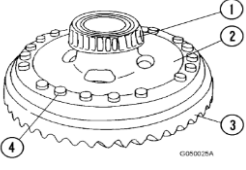
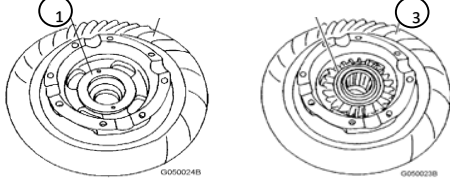
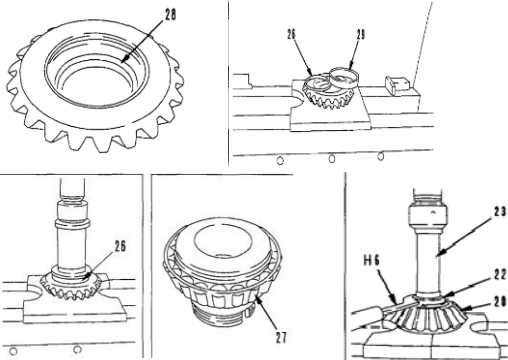
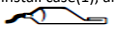
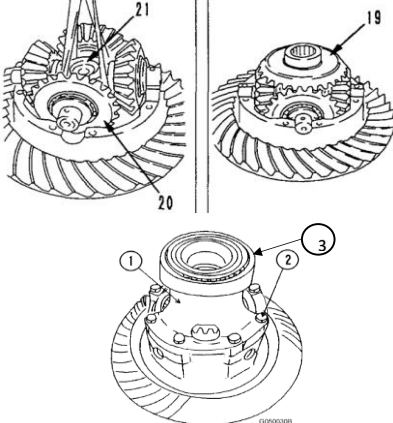
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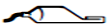
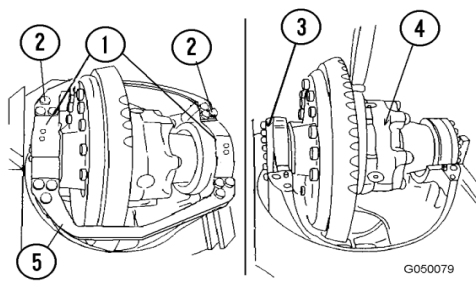
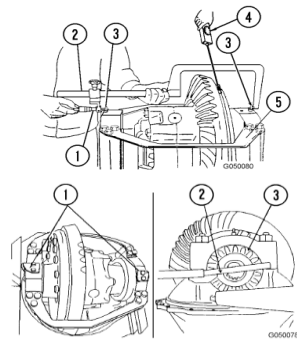
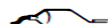
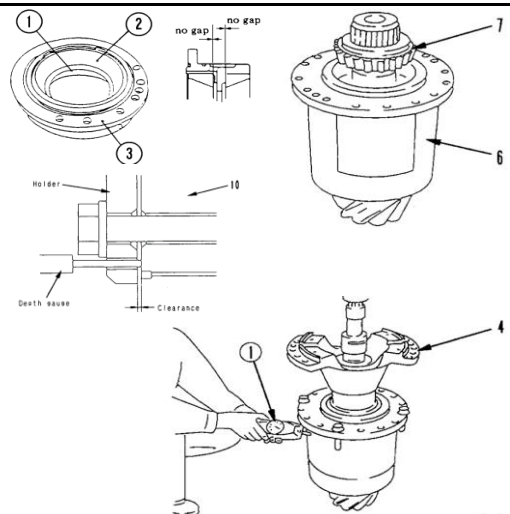
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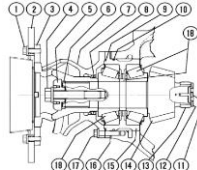
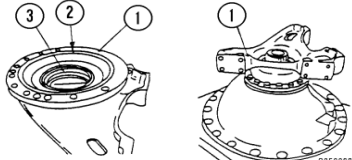
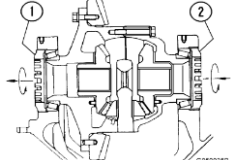
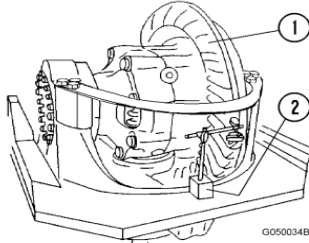
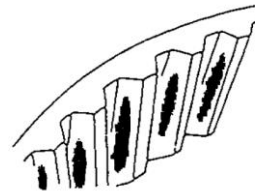
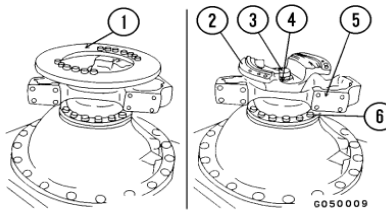
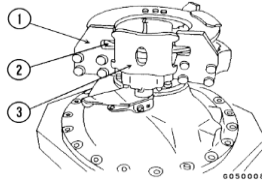
GL /QA

ASSEMBLY CHECK SHEET		WO NUMBER		DOCUMENT NO	RC/QR04/PT/15/02								
DIFFERENTIAL REAR		COMP.MODEL		PUBLIC DATE									
HD 1500-7		COMP.S/N		REVISION NO									
		UNIT MODEL	HD 1500-7	PAGE									
PROCESS		START		ASSEMBLY BY									
		FINISH											
NO	PART TO INSTALL OR ASSEMBLY	CONDITION TOBE INSTALL	MECH	GL/QA	SKETCH DRAWING OR PHOTHOES								
1	DIFFERENTIAL BEARING	Set the Differential case in tool repair stand. Press fit bearing in differential case(3) cooled -0° C with refrigerator) Install snap ring(2).											
2	DIFFERENTIAL GEAR ASSY	Align bevel gear(3) with dowel pins on differential gear case(2) and install. Apply thread tightener to THE CROWN(3) "NOT" CAPSCREW(4).  Adhesive (LT-2)  Tightening bolt <table border="1"> <tr> <th>Standard</th> <th>Actual</th> </tr> <tr> <td>134 Kgm</td> <td></td> </tr> </table> Install side bearing(1). Fit the bearing by heating 125° C(265 F)	Standard	Actual	134 Kgm				 				
Standard	Actual												
134 Kgm													
		Turn the case over. Align with dowel pins and install trust washer(1). Ensure the head of the dowel pin. <table border="1"> <tr> <th>Standard</th> <th>Actual</th> </tr> <tr> <td>0,5 +0,2 mm</td> <td></td> </tr> </table> Install side gear(3)	Standard	Actual	0,5 +0,2 mm								
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3	PINION GEAR ASSEMBLY	Install outer races (29) and (28). Press fit bearing (27) in shaft. Assemble collar (26). Use a press to install bearing. Assemble pinion gear (24) and bearing (25) in shaft. Fit lock plate and tighten lock nut (22) with wrench Bend the lock plate securely.  Tightening nut <table border="1"> <tr> <th>Standard</th> <th>Actual</th> </tr> <tr> <td>94,5 Kgm</td> <td></td> </tr> </table>	Standard	Actual	94,5 Kgm								
Standard	Actual												
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4	SPYDER GEAR ASSEMBLY	Assemble pinion gear assembly (20) in cross shaft (21), and install to case. Install side gear (19). Align with dowel pin & install thrust washer. Check that the head of the dowel pin <table border="1"> <tr> <th>Standard</th> <th>Actual</th> </tr> <tr> <td>0,5 +0,2 mm</td> <td></td> </tr> </table> Install side bearing(3). Fit the bearing by heating 125° C(265 F) Install case(1), and tighten mounting bolts(2).  Adhesive (LT-2)  Tightening bolt <table border="1"> <tr> <th>Standard</th> <th>Actual</th> </tr> <tr> <td>94,5 Kgm</td> <td></td> </tr> </table>	Standard	Actual	0,5 +0,2 mm		Standard	Actual	94,5 Kgm				
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		UNIT MODEL	PAGE	
HD 1500-7		HD 1500-7		

NO	PART TO INSTALL OR ASSEMBLY	CONDITION TOBE INSTALL	MECH	GL/QA	SKETCH DRAWING OR PHOTHOES																
5	CAP ASSEMBLY	Tighten side bearing adjusment nuts(3),and temporarily place differential gear in the case. Install cap(1) and tighten mounting bolts(2)  Adhesive (LT-2)  Tightening bolt <table><tr><th>Standard</th><th>Actual</th></tr><tr><td>175 Kgm</td><td></td></tr></table> Check the match mark on the bearing cap when installing. When tightening the bolts, rotate the bevel gear 20 - 30 times to settle the bearing before tightening.	Standard	Actual	175 Kgm				 G050079												
Standard	Actual																				
175 Kgm																					
6	PRELOAD ADJUSTMENT	Adjust the preload of the bevel gear as follows. 1) Install measurement adapters(2) to both ends of the cap. 2) Adjust with the nut so that the increase in dimention D is $0,66 \pm 0,12$ mm. <table><tr><th>Dimention</th></tr><tr><td></td></tr></table> 3)Fit spring balance to bevel gear and measure the starting force in the tangential direction should be:7,7 - 10,3 kg. <table><tr><th>Pre load Actual</th></tr><tr><td></td></tr></table> 4)Use of plastic mallet to seat bearings in carrier housing while turning crown wheel	Dimention		Pre load Actual				 G050080												
Dimention																					
Pre load Actual																					
7	PINION ASSEMBLY	Install bearing inner race(1) with tool heater to pinion gear(10) Install holder(1) and tighten mounting bolt.  Adhesive (LT-2)  Tightening bolt <table><tr><th>Standard</th><th>Actual</th></tr><tr><td>28 ± 3 Kgm</td><td></td></tr></table> Install outer race (1) & (2) to cage & check that there is no clearance between outer race and carrier Install pinion shaft to cage (6). Install bearing (7). Troughly coat the bearing rollers with oil Install coupling (4), and tighten holder with mounting bolt temporarily, then measure clearance between holder and pinion. <table><tr><th>Shim Actual</th></tr><tr><td></td></tr></table>  Adhesive (LT-2)  Tightening bolt <table><tr><th>Standard</th><th>Actual</th></tr><tr><td>280 ± 30 Kgm</td><td></td></tr></table> Rotate the cage 20 - 30 times, then using push-pull scale , check the starting torque. <table><tr><th>Starting Torque</th></tr><tr><td></td></tr><tr><th>Standard</th><th>Actual</th></tr><tr><td>< 5,1 Kg</td><td></td></tr></table>	Standard	Actual	28 ± 3 Kgm		Shim Actual		Standard	Actual	280 ± 30 Kgm		Starting Torque		Standard	Actual	< 5,1 Kg				
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ASSEMBLY CHECK SHEET DIFFERENTIAL REAR HD 1500-7	WO NUMBER	DOCUMENT NO	RC/QR04/PT/15/02
	COMP.MODEL	PUBLIC DATE	
	COMP.S/N	REVISION NO	
	UNIT MODEL	PAGE	

NO	PART TO INSTALL OR ASSEMBLY	CONDITION TOBE INSTALL	MECH	GL/QA	SKETCH DRAWING OR PHOTHOES
8	CAGE ASSEMBLY	Fit O-ring(10) and assemble shim(16), then install pinion and cage assembly to differential case. Shim Actual			
		Install oil seal and o ring to paking break support . Install support to differential case.then torque ,mounting bolt  Adhesive (LT-2) Tightening bolt Standard 56 ± 6 Kgm Actual			
9	BACKLASH AND TOOTH CONTACT ADJUSMENT	Adjust backlash as follows. 1)Move bevel gear with adjustment nut. ▲ IMPORTANT ▲ When adjusting the bevel gear, do not change the preload of the bearing. Always turn the adjustment nuts at both ends the same amount in the same direction.			
		2)Put dial gauge (2) at right angles in contact with the face of the tooth at the outside of the bevel gear. Turn the adjustment nut and adjust the backlash Backlash :adjust at 3 - 4 places and keep the pinion gear locked when measuring. Standard 0.46 - 0,66 mm Actual			
		3)Adjust tooth contact Coat face of 7 or 8 teeth of bevel pinion lightly with red lead (minimum). Hold the bevel gear by hand to brake it, and rotate the bevel pinion forward and backward and inspect the pattern left on the teeth. Centering the pattern between two sides contact surface			
10	CALIPER ASSEMBLY	Install Brake disc (1) and tighten mounting.  Adhesive (LT-2) Tightening bolt Standard 76 Kgm Actual			
		Install Plate (1) On one side ,temporarily. Fit pad(2) and install caliper(3). Install plate on the other side and tighten.  Adhesive (LT-2) Tightening bolt Standart 94,5 Kgm Actual			

ASSEMBLY CHECK SHEET DIFFERENTIAL REAR HD 1500-7	WO NUMBER		DOCUMENT NO	RC/QR04/PT/15/02
	COMP.MODEL		PUBLIC DATE	
	COMP.S/N		REVISION NO	
	UNIT MODEL	HD 1500-7	PAGE	

NO	PART TO INSTALL OR ASSEMBLY	CONDITION TO BE INSTALL	MECH	GL/QA	SKETCH DRAWING OR PHOTOS
11	CALIPER ASSEMBLY	<i>Note: Adjust clearance of 0,1 mm between the plate and caliper.</i> Install spring cylinder assy (1) and connect rod clevis with connection pin(3) to the lever of slack adjuster(4).			

NOTE

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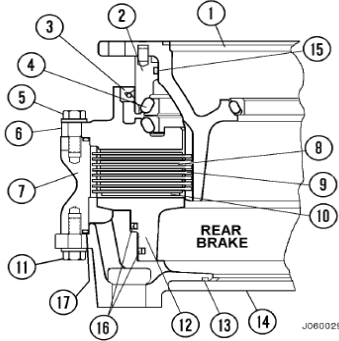
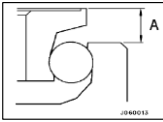
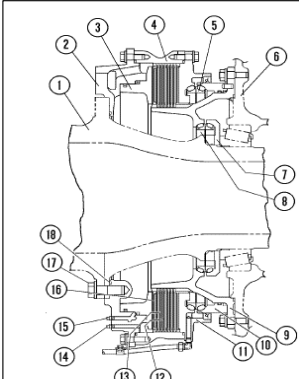
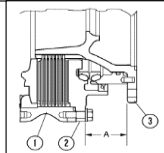
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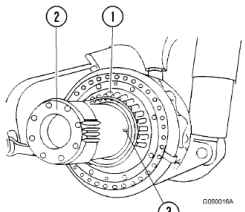
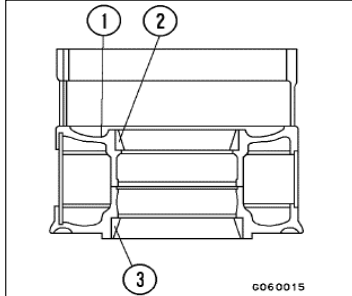
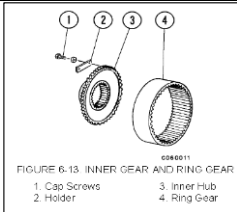
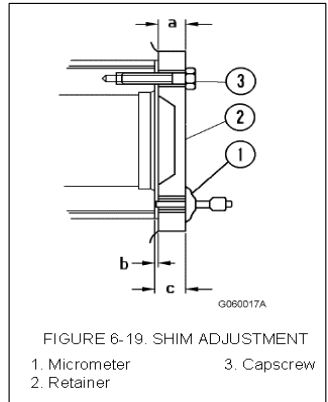
ASSEMBLY CHECK SHEET FINAL DRIVE HD 1500-7	WO NUMBER		DOCUMENT NO	
	COMP.MODEL		PUBLIC DATE	
	COMP.S/N		REVISION NO	
	UNIT MODEL		PAGE	

PROCESS	START		DISASSEMBLY BY	
	FINISH			

NO	PART TO REMOVE OR INSPECT	CONDITIONS TOBE INSPECTION	MECH	GL	SKETCH DRAWING OR PHOTOES	REMARK
1	REAR BRAKE ASSEMBLY	⚠ IMPORTANT ⚠ Clean all parts, and check for dirt or damage. Coat the sliding surfaces of all parts with hydraulic oil before installing. NOTE: When assembling the floating seals, use alcohol to remove all of the white powder from the Oring surface before assembling. • Install floating seal assembly (4). to seal carrier (2) and hub (6). • measure dimension "a" between the hub and floating seal ring at four places around the circumference. The measurements must be within 1 mm (0.039 in) of one another Actual measure NOTE: When assembling brake components, align individual parts according to match marks made during disassembly. • Position the cylinder on a work surface, as shown in Figure 6-7. Install new seals on piston Lubricate the seals and install the piston in cylinder (14). • Lubricate and install O-ring (17). • Install ring gear (7, Figure 6-7) onto cylinder housing (14). • Install capscrews and flatwashers (11) Tighten the bolt to 55 ± 5 Kgm Actual Torque:  FIGURE 6-7. BRAKE ASSEMBLY COMPONENTS 1. Inner Gear 2. Seal Carrier 3. Lip Seal 4. Floating Seal 5. Capscrew & Washer 6. Hub 7. Ring Gear 8. Separator Plate 9. Friction Disc 10. Damper 11. Capscrew & Washer 12. Piston 13. O-Ring 14. Cylinder 15. O-Ring 16. Piston Seals 17. O-Ring  FIGURE 6-8. SEAL POSITION CHECK 1. Ring Gear 2. Hub 2	REAR BRAKE INSTALLATION	The brake assembly weighs approximately 1100 kg (2425 lbs) • The brake housing must be properly oriented to align with the hydraulic connections and mounting holes on the inner side. • Place O-ring (18, Figure 6-5) into position and ensure the O-ring remains in place during installation. • Move the brake assembly into position against the flange on the axle. Coat mounting capscrew threads (17)with Three Bond (p/n TB1374 or 09940-00030). Install the Bolt and washers, and tighten to 172 ± 20 Kgm. Actual Torque: • Measure the distance between the inner gear in four places 90° apart. Dimension "A" should be 118 ± 1.0 mm (4.65 ± 0.04 in.) Actual Measure: A B C D  FIGURE 6-5. REAR DISC BRAKE ASSEMBLY 1. Rear Axle 2. Cylinder 3. Piston 4. Ring Gear 5. Floating Seal Assembly 6. Wheel Hub 7. Retainer 8. Floating Seal Assembly 9. Inner Gear 10. Seal Retainer 11. Hub 12. Separator Plate 13. Friction Disc 14. Housing Drain Plug 15. Piston Cavity Plug 16. Damper 17. Capscrews 18. O-Ring  FIGURE 6-6. SEAL POSITION CHECK 1. Ring Gear 2. Hub 3. Inner Gear		

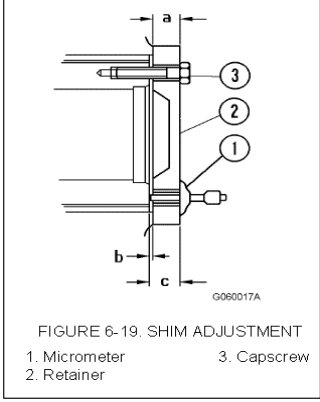
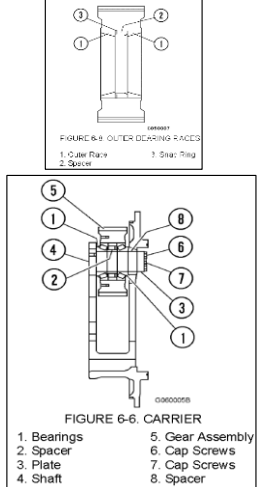
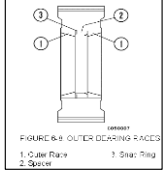
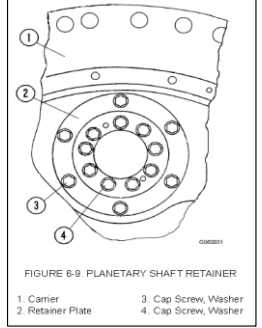
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NO	PART TO REMOVE OR INSPECT	CONDITIONS TO BE INSPECTION	MECH	GL	SKETCH DRAWING OR PHOTOES	REMARK
3	INNER BEARING	- Install floating seal assembly (13) to the wheel hub seal carrier. Install a new O-ring on the seal carrier. 2. Install inner bearing (1, Figure 6-18) and pin (3).			 FIGURE 6-18. BEARING 1. Bearing 2. Axle 3. Pin	
4	ASSEMBLY WHEEL HUB	- Install floating seal assembly to the wheel hub seal carrier. Install a new O-ring on the seal carrier. Use a bearing plate and press to install outer races (2 & 3, Figure 6-17) into wheel hub (1). - Raise the rear wheel hub assembly. Slide the assembly over the axle and against the disc brake inner gear. - Install the outer bearing retainer pin onto the axle. Install outer bearing d. Install nuts (2, Figure 6-15) and tighten to 93 Kgm. Actual Torque: <i>NOTE: DO NOT remove tools holder until the retainer has been completely secured.</i>			 FIGURE 6-17. HUB AND BEARING 1. Hub Assembly 2. Outer Race 3. Outer Race	
5	RING GEAR ASSY	a. Place inner hub (3, Figure 6-13) into position on ring gear (4). b. Install holder (2). Apply Three Bond #1374 thread tightener to mounting cap screws (6) and tighten to 113 N-m (83 lb ft). c. Install ring gear assembly (1, Figure 6-12).			 FIGURE 6-13. INNER GEAR AND RING GEAR 1. Cap Screws 2. Holder 3. Inner Hub 4. Ring Gear	
6	ADJUST BEARING PRELOAD	a. Install retainer (2, Figure 6-19) with four equally spaced cap screws (3). DO NOT install any shims at this time. b. Remove fixture tools (1, Figure 6-15) at three places. c. Rotate the wheel hub five to six times. - Tighten the bolt evenly and alternately to 17±1 Kgm (123 ± 7 lb ft). Actual Torque: ▲ IMPORTANT ▲ If the cap screws are tightened without rotating the wheel hub, the bearing will not seat properly and the correct preload cannot be obtained. - Using depth micrometer (1, Figure 6-19) measure dimension (c) between the end of the shaft and the outer edge of retainer (2). - Install tools (1) at three evenly spaced locations.			 FIGURE 6-19. SHIM ADJUSTMENT 1. Micrometer 2. Retainer 3. Capscrew	

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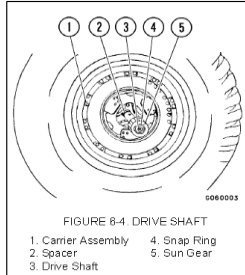
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NO	PART TO REMOVE OR INSPECT	CONDITIONS TO BE INSPECTION	MECH	GL	SKETCH DRAWING OR PHOTOES	REMARK
		- Remove the retainer, and measure thickness (a, Figure 6-19) of the retainer. Select a shim thickness which is equal to dimension (b). Use the formula below: $b = (c-a) + 0.3 \text{ mm (0.012 in.)}$ Actual Measure: - Insert the shim pack and install retainer (2, Figure 6-19). Apply Three Bond #1374 thread tightener to cap screws (3) and tighten to 94 Kgm. Actual Torque: - Remove tools (1, Figure 6-15) at three places. Install the absent cap screws (3) and tighten to 55 Kgm. Actual Torque:			 FIGURE 6-19. SHIM ADJUSTMENT 1. Micrometer 2. Retainer 3. Capscrew	
		Rotate the wheel hub five to six times. Tighten cap screws (2) evenly and alternately until the tightening torque is constant. After tightening the cap screws, rotate the wheel hub and verify smooth rotation.				
7	PLANETARY CARRIER ASSY	- Install spacer in the carrier. - Install snap ring (3, Figure 6-8). - Position spacer (2), and press fit outer race (1). - Set spacer (2, Figure 6-6) and bearing (1) into position. Raise the planetary gear and assemble. Install bearing (1) from the top. - Push planetary gear assembly (5) into the carrier. - Temporarily install plate (3) with cap screw (7). Install two guide cap screws to shaft (4), and press fit. - Coat Three Bond 1374 thread tightener to cap screws (4, Figure 6-9). Tighten to 11,5 Kgm. Actual Torque: - Rotate the pinion gear five to six revolutions. Again, tighten nine cap screws (4) and six capscrews (3) to the specified torque.			 FIGURE 6-6. CARRIER 1. Bearings 2. Spacer 3. Plate 4. Shaft 5. Gear Assembly 6. Cap Screws 7. Cap Screws 8. Spacer  FIGURE 6-8. OUTER BEARING FACES 1. Outer Race 2. Bearing 3. Snap Ring  FIGURE 6-9. PLANETARY SHAFT RETAINER 1. Carrier 2. Retainer Plate 3. Cap Screw, Washer 4. Cap Screw, Washer	



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8		- Lift the shaft into position. Install an eyebolt at the end of drive shaft (3). Insert a bar through the eyebolt and use the bar to push the drive shaft into the carrier. - Install spacer (2 Figure 6-4). - Install sun gear (5) on drive shaft (3). - Install snap ring (4). - Place cover (1) into position. Install the cap screws and washers that secure the cover.			 FIGURE 6-4 DRIVE SHAFT 1. Carrier Assembly 4. Snap Ring 2. Spacer 5. Sun Gear 3. Drive Shaft	

NOTE :

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